

MirrorFilter — Handoff

Current version: Web v5.227

Game patch: current with PoE2 0.5.5 (Forbidden Rites)

Live site:

<https://countinbiskitsyo.github.io/mirrorfilter/>

Repo: github.com/countinbiskitsyo/mirrorfilter
(index.html, README.md, LICENSE.txt)

Replaces the v5.205 handoff. A lot changed — see §3, §6 and §10.

1. WHAT THIS IS

A browser-based PoE2 loot filter editor. **One self-contained** `index.html`

(~1.8 MB: ~41k lines plus a 133 KB embedded base-item data table), deployed to

GitHub Pages. No build step. Upload the file, hard refresh, done.

Rollback = revert one file.

The user (Adamg) maintains a real 364-rule filter with it and plays the game

with the output. **The exported** `.filter` **is the only**

thing that reaches the
game — everything else is editor.

2. HOW TO WORK ON IT

The repo now carries **this handoff and the extractor as well as index.html**,
so a new session can pull everything it needs without
the user uploading
anything:

```
B=https://raw.githubusercontent.com/countin  
biskitsyo/mirrorfilter/main  
curl -sL -o index.html  
$B/index.html  
curl -sL -o MirrorFilter_Handoff.md  
$B/MirrorFilter_Handoff.md  
curl -sL -o extract_basedata.js  
$B/extract_basedata.js  
# ...edit...  
node -e "...parse both <script> blocks..."  
# ALWAYS validate  
# browser-test with playwright  
# bump line 3 marker AND CHANGELOG[0].v  
(they must match)  
cp index.html /mnt/user-  
data/outputs/index.html  
present_files
```

Check line 3's version against this doc. If it's newer, the user shipped something this doc doesn't know about — trust the file and ask.

Playwright is available (`npm install playwright`, chromium at `/opt/pw-browsers/chromium-1194/chrome-linux/chrome`, use `--no-sandbox`).

Sandbox network is restricted to package registries + GitHub. poe.ninja, poe2db and the Cloudflare Worker are unreachable. Economy features and anything fetching base data can only be verified by the user.

Export-parity test — run after ANY change

The single best proof that editor churn didn't change game output:

```
await pg.evaluate(()=>loadExampleFilter());
const g = generateFilter(S,'T','T');    //
returns a TUPLE; the text is g[0]
```

Expect 364 rules, 291 Show, 3 Hide, 0 bare BaseType lines, 195,509 chars.
Compare byte-for-byte against the previous build.

Dismiss BOTH first-run prompts in every test

```

await
pg.evaluate(()=>document.getElementById('_b
kNever')?.click());
await pg.evaluate(()=>{const b=
[...document.querySelectorAll('button')]
    .find(x=>/No
thanks/i.test(x.textContent));
b&&b.click();});

```

Other test gotchas

- Hidden elements return empty `innerText` — use `textContent` .
- `page.click('text=...')` collides with repeated labels and gets blocked by the tooltip layer. Call the handler via `page.evaluate` instead.
- `parseFilterText()` returns `section_tree` (snake_case), not `sectionTree` . Assigning its result wholesale onto `S` silently leaves the tree empty.
- The tab is `viz` , not `stats` (`switchTab('viz')`).
- There is no `renderAll()` . The refresh idiom is `saveState(); renderList(); renderEditor(byId(activeId));`

3. ARCHITECTURE LANDMARKS

Two `<script>` blocks

- **Block 0** — engine: parser, `generateFilter`, `buildMeta`, the backend shim IIFE (IndexedDB, File System Access), `window.MF_WEB` bridge.
- **Block 1** — UI: `S` state, rendering, all panels, `CHANGELOG`, `__MF_BASEDATA__`, `__MF_DEFAULT_TODO__`.

A top-level `const` in a classic script is shared across blocks, so block 0

functions CAN see block-1 constants at call time. But anything declared *inside*

the block-0 IIFE (e.g. `_dirHandle`) is invisible from outside — route those through `window.MF_WEB`.

Key globals

- `S` — the whole editor state. Persisted to IndexedDB via `saveState()`.
- `ALL_BASETYPES = CSUGG_CURRENCY_POE2 + CSUGG_BASETYPES_POE2` (3,098).

The database is in TWO halves — always use

`ALL_BASETYPES`. It feeds three things: BaseType autocomplete, paste validation, and the unknown-BaseType check.

- `__MF_BASEDATA__` – **generated, do not hand-edit** (see §6).
- `LEAGUE_TIMELINE` – single source of truth for leagues (§5).

`saveState()` re-reads the DOM

It overwrites `S.filterName` and `S.league` from the input and dropdown. Code that sets those then calls `saveState()` must push the values into the controls
`first ( syncRestoredIdentityToDOM() )`.

Section ids vs display paths

A rule's `.section` holds a section **id**. Imported Magic/Normal sections often carry auto-generated `sec_XXX` ids while Rare ones carry readable path ids, so **ids cannot be compared across trees**. Use `secPath(id)` for the display breadcrumb (segments joined by `>`). The mirror feature depends on this.

4. LOAD-BEARING SAFETY SYSTEMS – DO NOT REGRESS

1. **Empty condition lists emit nothing.** An empty `BaseType` array used to write a bare `BaseType ==`, silently widening the rule to whatever remained. Guard is in `generateFilter`.
 2. **Rules that lose all BaseTypes to league filtering are commented out** with `[NO CONTENT THIS LEAGUE]`.
 3. **Section identity (@id: tags)** keeps same-named sections apart.
 4. `healSectionIdentity` must stay a no-op on clean data. Test it.
 5. **Auto-Tier collision guard.**
 6. **An unrecognised BaseType stops PoE2 loading the filter entirely** — not a skipped line, no filter at all.
 7. **Mirror previews before writing** and re-derives its plan on Apply, refusing if the filter changed since the preview was drawn.
 8. **Tooltips never invent data.** A base absent from `__MF_BASEDATA__` shows no stats at all. Keys are omitted rather than zero-filled, so "none" and "unknown" stay distinguishable.
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5. LEAGUE SYSTEM

Adding a league = ONE line. Append to

`LEAGUE_TIMELINE` (oldest first).

Name must match poe.ninja exactly. **Leave old leagues in.**

Hardcore is a checkbox; `composeLeague()` handles hardcore Standard being just `"Hardcore"`.

League Content records only *exceptions* and **fails open everywhere.**

Currently seeded: Sacred Bloom: { addedIn: 'Forbidden Rites' } .

6. BASE-ITEM DATA (tooltips) — GENERATED, DO NOT HAND-EDIT

`window.__MF_BASEDATA__` holds **1,744 bases across all 29 classes**, powering the in-game-style tooltip shown when hovering a BaseType chip or an Economy Prices row.

Source: poe2db.tw, data-mined from the game client (the same source poe2filter.com uses for exact names). **Generated by** `extract_basedata.js`, which is embedded in `index.html`, downloadable from

Resources -> Maintenance & Data

-> Base Item Data, AND committed to the repo. It cannot be lost.

Refresh after a patch: the Base Item Data panel flags itself "needs rebuilding" when `BASEDATA_PATCH` no longer matches `GAME_PATCH_BASELINE`.

Re-run the script, diff the new `basedata.js` against the old, paste over the

`__MF_BASEDATA__` block, **and update**

`BASEDATA_PATCH` / `BASEDATA_GENERATED`

or the panel keeps reporting the old build. The diff is the list

of what GGG changed to item bases, which also matters for stat-based rules

(see §7 T2).

Schema — a key is ABSENT when the base lacks that stat:

`b` =block% `a` =armour `e` =evasion `s` =energyShield
`w` =runicWard

`pd` =physDamage `c` =crit% `aps` =attacksPerSec

`rg` =range `p` =[other props]

`lv` =level `st` / `dx` / `it` =attribute reqs `im` =implicit

Verification that matters: Sectarian Crest Shield reads 61 Armour / 19 ES / Level 33 / 27 Str / 27 Int in the table, on poe2db, and on the user's real in-game item. All 143 shields from the

earlier hand-made table
agree with the generated one exactly.

Writing a poe2db parser — read this before touching it

It took SIX attempts, because the page was parsed by imagination rather than by evidence. **The stat LABEL is itself a hyperlink:**

```
<div class="property"><a ...>Armour</a>:  
<span>61</span></div>  
<div class="property"><a ...>Block</a>  
chance: <span>25%</span></div>
```

Flattening the page to text therefore ALWAYS separates a label from its value, and splits "Block chance" in half. Line-based parsing cannot work. Parse the HTML structure per item card, stripping tags only *within* one `property` div. Base items carry `class="whiteitem"`; uniques carry `class="UniqueItem"` — that attribute is what keeps the two apart.

If it ever needs changing, get the real HTML first (`--raw --debug` dumps the full page) and test against that, never against an invented sample.

7. OPEN ITEMS

T1 – Sacred Bloom's Class is UNCONFIRMED (user's task)

Placed in **Omens** (`Class: Omen`). Sources disagree: poe.ninja files it under

`/omens/` , Game8 calls it a Currency, FilterBlade groups it with map fragments.

A wrong Class fails SILENTLY – the rule needs Class AND BaseType, so it simply never matches. Only an in-game hover settles it.

CONFIRMED IN GAME – "no defence" is a usable condition

PoE2 evaluates `BaseArmour <= 0 /`

`BaseEnergyShield <= 0 / BaseEvasion <= 0`

as TRUE for a base that has none of that stat (verified live with quivers,

which have no defences whatsoever). So the seventh stat combination – all

three at `<= 0` – is a real, general rule and not a trick.

The user added

Ward / No Defence Based to Off-hands with exactly that, and quivers now label correctly in game.

T2 – The same gap still exists in the ARMOUR section

Off-hands is solved. Armour is not. Counting against

`__MF_BASEDATA__`:

Section	Bases with NO armour/evasion/ES
Off-hands	31 (11 quivers, 18 ward-only, 2 odd shields) – FIXED
Armour	70 (68 ward-only Runeforged, 2 others) – still falling through

Fix is the same seventh rule, with the armour classes:

`Class == "Body Armours" "Boots" "Gloves"`

`"Helmets"` plus the three `<= 0`

conditions. Build it in Rare, then use Mirror with **"Also create missing**

rules and sections" ticked to generate the Magic and Normal copies.

T2b – Ward-only shields (now covered, left here as context)

The user converted Rare/Magic/Normal Off-hands from BaseType lists to

`BaseArmour / BaseEnergyShield / BaseEvasion`

conditions – good, and

self-maintaining. But **13 high-tier**

Runeforged/Runemastered shields have ALL defence converted to Runic Ward, with zero Armour, Evasion and Energy Shield, so they match none of the three rules and hit **Fallthrough!**. Examples: Runeforged Stoic Crest Shield (Ward 146), Runeforged Grand Targe (176), Runemastered Crucible Tower Shield (250). Their tooltips now say so explicitly. Open question: their own rule, or folded into the existing three?

Known, documented, not fixed

- **1b ghost sections** — the *bundled example filter* gains 5 empty sections on round trip (86 → 91). Converges, harmless, **does not affect the user's real filter** (88 → 88).
- **BaseType database gaps** — 499 equipment bases were added in 5.211 (2,599 → 3,098), but ~497 Runeforged and ~180 Runemastered variants are still missing, and currency/gems/waystones/omens/essences were never audited. If filled, the BaseType check could become a hard pre-export gate.

Cannot be done from the sandbox

Price accuracy, Worker health, poe2db fetches.

8. THINGS THAT BIT US (don't repeat)

- **Writing a parser against an imagined page.** Six failed versions; the fix was asking for raw HTML. See §6.
- **A crashing build script truncated index.html to 0 bytes,** and the empty file was copied to outputs before anyone noticed. Recovery came from the user's GitHub upload. Two habits that follow: never pipe a whole-file rewrite through something that can die mid-write without a copy in hand, and check the staged file's SIZE and version line before presenting it.
- **Declaring a variable in the wrong function** blanked the entire Visuals & Stats left column. The suite passed because no test opened that tab.
The user found it, not the tests.
- **Editing `__MF_DEFAULT_TODO__` reached nobody.** The built-in list was only

used when the user had NO saved list — i.e. never, for any real user. Now a `seedIntoExisting` flag merges new items once, and delivered ids are recorded so a deleted item stays deleted.

- **One shared folder handle.** Export/Quick-Save, every Multi-Location Save entry and disk auto-backup all share `_dirHandle`. Picking a folder for one silently re-points all of them — which is how Quick-Save stopped writing to the game folder. 5.207 added a confirm and a staleness flag; the one-slot design is unchanged.
- **Centred toolbar + variable-width children = visible shifting.** The status text and the pulsing Auto-Tier border changed width mid-action and slid the whole bar. Reserve fixed slots.
- **Free prose in a `#` comment next to a rule** was parsed as a section header on re-import, inventing a phantom section.
- **Comma-splitting pasted text** once corrupted the database.
- **Curly apostrophes** from wiki copy-paste don't match the database; `parseValueList` normalises them.

- `doNewFilter` didn't clear `sectionTree` — exported 448 lines of empty headers.
 - **Guides/README drifting behind the tool.** When behaviour changes, grep them.
 - **Inline `onclick` with nested quotes** leaked raw JS into visible text.
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9. TESTING NOTES

Verified against the real filter:

- import → 364 rules, 88 sections, 0 orphans
- export → Show 286, Hide 3, 0 malformed lines (bundled example: Show 291)
- round trip ×3 → stable at 364/88
- `generateFilter` with an empty `BaseType` array → no bare line
- `healSectionIdentity(S)` → `changed=false`
- Visuals & Stats → both columns render (catches the scope bug)
- no scrollbar at 2560×1222
- `scrollToSaveStage()` → lands on Resources, no console errors
- mirror preview on the real filter → exactly 10 changes, idempotent on rerun

- `buildBaseTip('Chaos Orb')` → `' '` (never invents data)
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10. WHAT CHANGED SINCE v5.205

- **5.207** header-shift fix; shared-folder overwrite guard + stale-location flag
- **5.208–5.210** welcome box rebuilt as the same ①②③ stages used in Resources; Saving Your Work simplified and colour-themed per stage
- **5.211** 499 missing equipment bases added to the BaseType database; export verified byte-identical
- **5.213** Test Item can evaluate `BaseArmour / BaseEnergyShield / BaseEvasion` (it never read those lines before), and is honest about augmented values
- **5.214** to-do seeding actually reaches an existing list
- **5.215 Mirror Rare** → **Magic/Normal** (Resources → Filter Cleanup Tools)
- **5.217** mirror preview legend
- **5.218–5.223** BaseType tooltips: shields-only hand table → 1,744 bases across all 29 classes, generated and verified

- **5.225** to-dos for tooltips phase 2, patch-refresh, and moving the base-data refresh into the tool via the Worker
 - **5.227** Mirror can create missing rules AND sections (checkbox, off by default). New rules take Rarity/AreaLevel/visuals from a DONOR rule already in the target tree, never from the Rare rule — otherwise a new Magic rule would arrive as Rarity Rare wearing rare colours. No donor = blocked and reported, never guessed.
 - **5.226** Resources -> Maintenance & Data -> **Base Item Data** panel: the extractor is embedded in index.html and downloadable, the data's build patch is compared against the game patch with a "needs rebuilding" badge, and the rebuild steps plus a copy-the-message-for-Claude button live in the panel
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11. USER PREFERENCES

- Wants to know *why*, not just *what* — explain reasoning and trade-offs.
- Values honesty about uncertainty over false confidence. Say when something is

untested or unverifiable — **up front, not after it fails.**

- Cares a lot about not breaking a working tool. Offer rollback, keep backups, measure before and after.
- Dislikes wordiness in UI copy. Short lines, bold only what matters.
- Wants UI that is **scannable by label**, not prose read top-to-bottom.
- Will run scripts and paste results back — a good channel for anything the sandbox cannot reach. Give printable instructions when asked.
- **The backup system is "the safety net of the entire project"** — err toward more explanation there, never less.